1.2

A Parallel Comparison between Human Brain and AI Agents

人脑与 AI 智能体的平行比较

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Advances and Challenges in Foundation Agents:
From Brain-Inspired Intelligence to Evolutionary, Collaborative, and Safe Systems

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> 原文中的表 1.1、图 1.1 和参考文献编号 [12]-[31] 均予以保留。引用编号对应的完整书目信息请查阅原论文参考文献表。中文为忠实阅读译文；术语或表述如有歧义，以英文原文为准。

文档结构

第一部分：英文原文摘录（保留原版排版、表格与插图）。
第二部分：中文译文（按内容结构重排，表格与图示标签另附中文对照）。

PART I

English Original Excerpt

英文原文摘录

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1.2 A Parallel Comparison between Human Brain and AI Agents

The rapid integration of LLMs into intelligent agent architectures has not only propelled artificial intelligence forward but also highlighted fundamental differences between AI systems and human cognition. As illustrated briefly in Table 1.1, LLM-powered agents differ significantly from human cognition across dimensions such as underlying "hardware", consciousness, learning mechanisms, creativity,

Section 1.2 begins here and continues on the next source page.

and energy efficiency. However, it is important to emphasize that this comparison provides only a high-level snapshot rather than an exhaustive depiction. Human intelligence possesses many nuanced characteristics not captured here, while AI agents also exhibit distinct features beyond this concise comparison.

Human intelligence operates on biological hardware, the brain, that demonstrates extraordinary energy efficiency, enabling lifelong learning, inference, and adaptive decision-making with minimal metabolic costs. In contrast, current AI systems require substantial computational power, resulting in significantly higher energy consumption for comparable cognitive tasks. Recognizing this performance gap emphasizes energy efficiency as a critical frontier for future AI research.

Human learning is continuous, interactive, and context-sensitive, deeply shaped by social, cultural, and experiential factors. Conversely, LLM agents primarily undergo static, offline batch training with limited ongoing adaptation capabilities. Despite research work using instruction tuning and reinforcement learning from human feedback (RLHF) [12], LLM agents still fall short of human-like flexibility. Bridging this gap through approaches such as lifelong learning, personalized adaptation, and interactive fine-tuning represents a promising research direction, enabling AI to better mirror human adaptability and responsiveness.

Creativity in humans emerges from a rich interplay of personal experiences, emotional insights, and spontaneous cross-domain associations. Emotional states not only motivate creative expression but also influence the originality, depth, and resonance of the outcomes, imbuing them with personal meaning and affective significance. In contrast, creativity in large language models (LLMs) stems primarily from the statistical recombination of training data (what might be described as "statistical creativity"). While often fluent and occasionally surprising, this form of creativity lacks emotional grounding, lived experience, and intentional originality. This contrast reveals opportunities for advancing AI agents by incorporating deeper contextual understanding, simulated emotional states, and experiential memory. Such developments could lead to more authentic and emotionally attuned creative processes.

In terms of consciousness and emotional experience, LLM agents lack genuine subjective states and self-awareness inherent to human cognition. Although fully replicating human-like consciousness in AI may not be necessary or even desirable, appreciating the profound role emotions and subjective experiences play in human reasoning, motivation, ethical judgments, and social interactions can guide research toward creating AI that is more aligned, trustworthy, and socially beneficial.

Considering the time scale, the human brain has evolved over millions of years, achieving remarkable efficiency, adaptability, and creativity through natural selection and environmental interactions. In stark contrast, AI agents have undergone rapid yet comparatively brief development over roughly 80 years since the advent of early computational machines. This parallel comparison between human cognition and AI systems is thus highly valuable, as it uncovers fundamental differences and provides meaningful insights that can guide advancements in AI agent technologies. Ultimately, drawing inspiration from human intelligence can enhance AI capabilities, benefiting humanity across diverse applications from healthcare and education to sustainability and beyond.

1.2.1 Brain Functionalities and AI Parallels

Designing intelligent agents calls for inspiration from the human brain's functional architecture. A high-level map linking brain regions – frontal, parietal, occipital, temporal lobes, as well as the cerebellum, brainstem, and key subcortical structures – to cognitive functions can reveal gaps between human capabilities and current AI systems, as shown in Figure 1.1. However, brain functions are not siloed into rigid anatomical zones: most abilities emerge from networks spanning multiple regions. For instance, memory involves the hippocampus (temporal lobe) interacting with frontal cortex and other areas, and "self-awareness" or consciousness cannot be pinpointed to a single spot. Therefore, it is important to keep in mind that cognition is distributed rather than strictly localized. With that in mind, we review each major brain region's core functions (drawing on *Principles of Neural Science* by Kandel et al. [13], *Neuroscience* by Purves et al. [14],

Table 1.1: Concise high-level comparison between human brain and LLM agent.

Dimension	Human Brain / Cognition	LLM Agent	Remarks
Hardware & Maintenance	- Biological neurons, neurotransmitters, neuroplasticity. - Requires sleep, nutrition, rest. - Limited replication, knowledge transfer via learning. - Extremely energy-efficient (approx. 20W).	- Deep neural networks, gradient-based optimization. - Requires hardware, stable power, and cooling. - Easily duplicated across servers globally. - High energy consumption (thousands of watts per GPU server).	Human brains are biologically maintained, energy-efficient, and not easily replicable. LLM agents rely on hardware maintenance, are highly replicable, but significantly less energy-efficient.
Learning Style	- Lifelong, continuous, online learning. - Few-shot, rapid knowledge transfer. - Influenced by environment, culture, emotions.	- Primarily offline, batch-based training. - Limited online learning and adaptation. - Neutral, impersonal learned knowledge.	Despite improvements via instruction tuning, human learning remains more dynamic, adaptive, and culturally/emotionally integrated than LLM learning.
Creativity & Divergence	- Rooted in personal experience, emotions, subconscious insights. - Rich cross-domain associations, metaphorical thinking. - Emotional depth influences creativity.	- Statistical recombination from extensive data. - Novelty through probabilistic optimization. - Limited emotional and experiential grounding.	LLM creativity is statistical and data-driven; human creativity blends emotion, experience, and subconscious processes.
Consciousness & Development	- Genuine subjective experiences, emotions, self-awareness. - Gradual developmental stages from childhood. - Emotional cognition drives decision-making.	- No genuine subjective experience or self-awareness. - "Emotions" are superficial language imitations. - Static post-training with limited dynamic growth.	Human consciousness emerges from emotional, social, and biological development; LLMs remain static without true introspection or emotional depth.

and other sources) and map them to AI-relevant cognitive capabilities. We then propose a set of functions to feature in the figure – emphasizing those most relevant to AI agents (e.g. reasoning, memory, planning, perception, decision-making, motivation, emotion, motor skills) – along with an assessment of how developed these functions are in AI. For a big-picture perspective, we categorize the state of research in AI with three distinct levels:

- **Level 1 (L1):** well-developed.
- **Level 2 (L2):** partially developed.
- **Level 3 (L3):** underexplored.

Our goal is to come up with a clear, biologically-grounded illustration and discussion that will engage researchers in AI by highlighting which human cognitive functions are replicated in machines and which remain frontier challenges.

Frontal Lobes (Executive Functions and Decision-Making) The frontal lobes – especially the prefrontal cortex – are the seat of the brain's highest-order cognitive functions known collectively as executive functions [13]. These include abilities such as planning, decision-making, problem-solving, working memory, and inhibitory control (self-control). The frontal lobe is also involved in voluntary motor control (with the rear portion containing the primary motor cortex) and aspects of language (Broca's area in the left frontal lobe

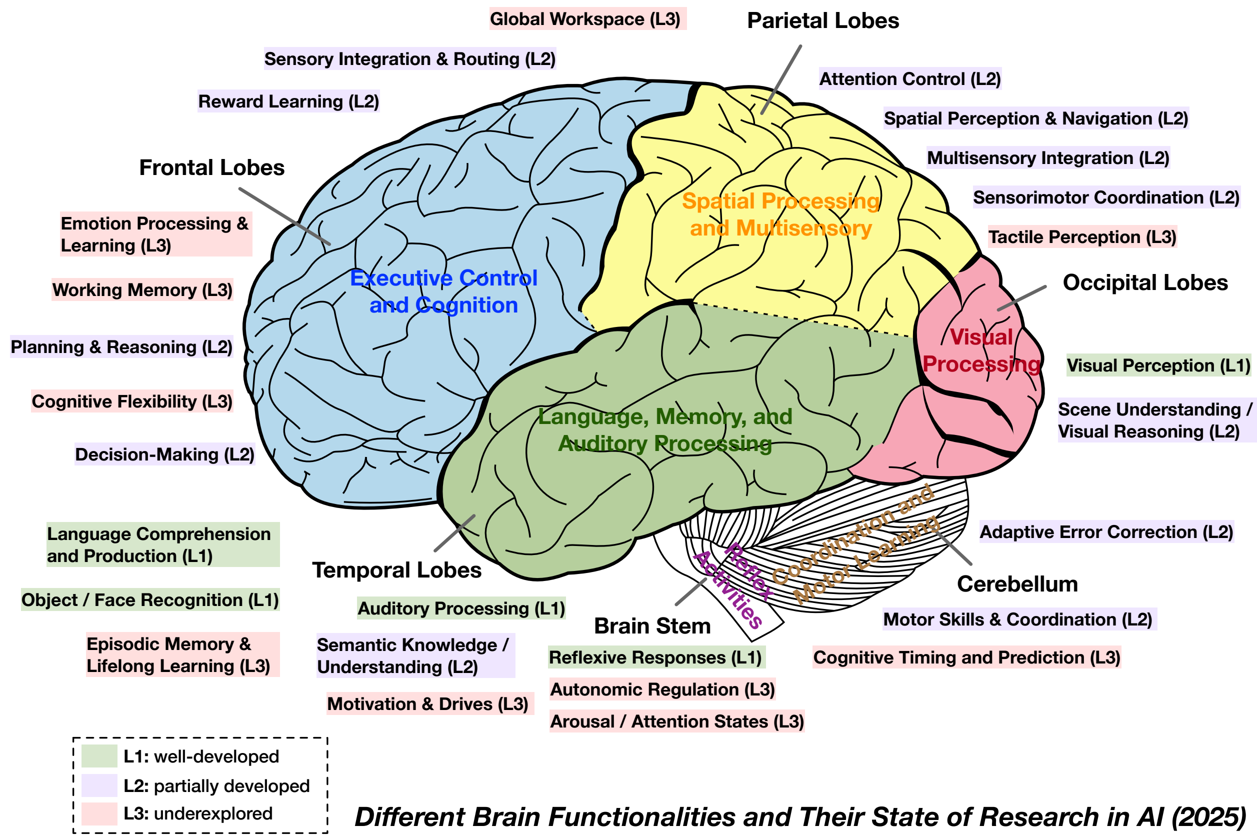


Figure 1.1: Illustration of key human brain functionalities grouped by major brain regions, annotated according to their current exploration level in AI research. This figure highlights existing achievements, gaps, and potential opportunities for advancing artificial intelligence toward more comprehensive, brain-inspired capabilities. Please note: The location of a label does not imply that it is associated with or is only associated with nearby brain regions. Also, to focus on the current state of AI capabilities rather than the anatomy of the brain, we did not mark the limbic system, subcortical system, hypothalamus, amygdala, and so on in this lateral view of the brain, though these systems play important roles in brain function.

handles speech production). From a neuroscience perspective, damage to the prefrontal cortex famously impairs one's judgment, planning, and social behavior (as illustrated by the classic Phineas Gage case) [15]. In the context of AI agents, frontal lobe functions correspond to the core "thinking" and control components of an intelligent system:

- **Planning and reasoning:** AI has made progress here, for example with automated planners and logical reasoners, and large language models (LLMs) that can follow multi-step reasoning to some extent. These are partially developed (L2) in current AI. However, human-level flexible planning remains only partly solved.
- **Decision-making:** In humans this involves weighing outcomes, rewards, and risks (frontal cortex often working with basal ganglia) [16]. AI agents have decision modules (e.g. reinforcement learning policies, decision trees, or LLMs), but handling open-ended, goal-conflicting decisions with human-like adaptability is still L2 at best. Simple decisions from well-defined rewards (like games) are mastered by AI, but broad autonomous decision-making in the real world remains challenging.
- **Working memory:** Frontal networks (especially dorsolateral prefrontal cortex) can hold and manipulate information in mind (e.g. remembering a phone number or interim result). AI analogs include the context windows of neural networks or explicit memory buffers. While current models

have limited memory (e.g. a Transformer's context length or external memory in some architectures), this is an active area, and partial functionality exists. Still, the robust, general working memory humans exhibit (flexibly updating and focusing on relevant info) is not fully realized in AI (some aspects may be underexplored, edging into L3).

- **Cognitive flexibility and inhibitory control:** Frontal lobes allow us to shift strategies or perspectives and to suppress inappropriate impulses. AI systems are typically brittle in this regard – they follow their programming or learned policy rigidly, and struggle with on-the-fly strategy shifts or inhibiting a pre-potent response unless explicitly trained. This remains underexplored (L3). For instance, an AI might exploit a reward loophole (lack of "inhibitory" self-regulation) unless designers anticipate and constrain it. Future agent architectures may need a mechanism akin to frontal inhibitory control to moderate behaviors.

It's worth noting that social-emotional functions involving the frontal lobe (like empathy, theory of mind, self-reflection) are still rudimentary in AI. Humans rely on frontal cortex (especially medial and orbitofrontal regions) interacting with limbic structures to navigate social situations and emotions. It is a network-level phenomena rather than confined to frontal lobe alone. AI agents do not yet possess genuine empathy or self-awareness – these functions remain L3 (largely absent). In summary, the frontal lobes contribute the supervisory, goal-directed intelligence that we often associate with "thinking", and many of these capacities are only partially realized in AI to date.

Parietal Lobes (Perception Integration and Attention) The parietal lobes are key to integrating sensory information from various modalities and constructing a spatial understanding of the world [14]. The anterior part of the parietal lobe contains the somatosensory cortex, which processes touch, proprioception (body position), and other somatic senses. The posterior parietal areas are crucial for spatial awareness, visuo-spatial processing, and attention – essentially, knowing where things are and how to interact with them. For example, the parietal lobe helps us localize objects in space, understand geometric relationships, and coordinate eye and hand movements by linking vision with motor plans. It also plays a central role in attention control, particularly the dorsal attention network that directs our focus to locations or sensory features of interest.

In AI terms, parietal lobe functions translate to an agent's ability to perceive and navigate its environment:

- **Multisensory integration:** Robots and AI systems that use multiple sensors (vision, touch, etc.) attempt to combine those inputs into a coherent model of the environment. This is still partially developed (L2) – e.g., we have AI that can align vision with depth sensors or touch, but human-level integration (where a slight brush on the arm, a sound, and a peripheral visual cue all unify into a single event perception) is far from achieved.
- **Spatial representation and mapping:** Parietal circuits create internal maps (for instance, of your surroundings or your body in space). AI has made progress in spatial mapping and navigation (SLAM algorithms for robots build 3D maps, and deep reinforcement learning agents can navigate virtual mazes). This capability is moderately developed – certain tasks like autonomous driving or drone flight show that machines can handle spatial reasoning in constrained scenarios (L2). Yet, they lack the general-purpose, flexible spatial understanding humans have (e.g. understanding a cluttered room's layout at a glance, or mentally rotating objects), so further research is needed.
- **Attention mechanisms:** The way the brain's parietal–frontal circuits spotlight task-relevant information is loosely echoed by the *attention heads* in modern transformer networks [17] and various attention mechanisms [18]. Humans, however, wield attention as an *active lens*: in a lecture we can *simultaneously* follow the speaker's voice (auditory stream), skim the projected slide (visual stream), and monitor the clock in peripheral vision, then "zoom in" on a line of text to decode a formula, or "zoom out" to grasp the talk's overall structure. Neurophysiology shows that such rapid shifts are

driven by top-down signals from prefrontal cortex and thalamic relays that modulate sensory gain on the fly [19]. By contrast, transformer attention is fixed once its weights are learned; it does not receive real-time executive feedback about goals or context. Hence we label current AI as L2: computational attention is powerful but still lacks the adaptive, goal-directed control that characterises biological attention, making this an active frontier of research.

- **Sensorimotor coordination:** Parietal lobe helps translate between sensory coordinates and motor coordinates – for instance, computing how to reach for a seen object (integrating visual location with arm position). Some AI systems (robotic manipulators with vision) approximate this, using calibration and learned coordinate transforms. Still, human parietal cortex excels at online adjustments and using context (like adjusting reach if an object is moving). AI is catching up in domains like robotic arm manipulation, but general sensorimotor integration remains L2 (demonstrated in specific setups but not as universally robust as in humans).

Notably, AI has very limited touch sensing (making that L3), whereas the parietal somatosensory cortex finely discriminates texture, pressure, etc. Overall, parietal lobe functions are critical for an agent to perceive its world and orient within it, areas where AI has some successes (especially in vision) but still lacks the generality and fluidity of human perception.

Occipital Lobes (Visual Processing) The occipital lobes are the brain's visual processing center [13]. The primary visual cortex (V1) in occipital lobe receives input from the eyes (via the thalamus) and extracts low-level features like edges. From there, occipital regions and adjacent visual areas (extending along the occipital-temporal border for the ventral stream, and occipital-parietal for the dorsal stream) hierarchically build up visual perception: detecting shapes, colors, motion, and eventually complex patterns and objects. In summary, the occipital lobe is primarily responsible for processing visual information.

In the context of AI, vision has been one of the most successful domains – thanks largely to deep learning:

- **Visual perception (Recognition):** Machine vision systems (convolutional neural networks and their successors) can now match or exceed human performance in tasks like object recognition, face detection, and image classification [20]. This corresponds to the L1 level (well-developed) in AI. For example, AI vision models can instantly recognize thousands of object categories in images, a feat once thought to require human-like vision. This maps to what the occipital lobe and ventral visual cortex do – identifying what is in the visual field.
- **Scene understanding and visual reasoning:** Beyond raw perception, humans readily understand spatial relationships in a scene, contextual clues, and can perform reasoning on visual inputs (e.g. predicting what might happen next in a scene, or solving a visual puzzle). AI is part-way (L2) here. Some systems can caption images or answer questions about a scene (vision-language models), indicating a degree of semantic understanding. Yet, these models often lack true grounded understanding – they might label objects but fail on deeper comprehension (for instance, understanding intentionality or causality from an image). Visual reasoning tasks (like answering abstract questions about a picture or performing complex video analysis) remain challenging.
- **Visual attention and eye movements:** Humans constantly move their eyes and focus on important parts of the visual field (a function involving occipital and parietal circuits). AI vision models don't literally move eyes, but some incorporate attention mechanisms that mimic focusing on regions of an image. This is related to the earlier discussion on attention (shared with parietal function). It's moderately well implemented in AI (L2), but we don't explicitly label this under occipital as it can be considered part of the general attention function under parietal/frontal coordination.

AI has heavily developed the visual recognition capabilities, but capabilities like real-time 3D visual guidance are less mature (though present in robotics and self-driving cars). This distinction shows another gap in AI relative to the human visual system's integrated prowess.

Temporal Lobes (Memory, Language, and Audition) The temporal lobes have diverse but crucial roles in cognition, spanning auditory processing, language, memory, and high-level visual recognition [14]. The upper part of the temporal lobe (superior temporal gyrus) contains the primary auditory cortex, which processes sound inputs – frequencies, rhythms, etc. Adjacent areas (e.g. Wernicke's area in the left temporal lobe [21]) are essential for language comprehension, linking sounds to meaning. The medial temporal lobe houses the hippocampus and related structures, which are the heart of the brain's episodic memory system (forming and retrieving autobiographical memories) and also support spatial navigation. The temporal lobe's ventral visual stream (inferior temporal cortex) specializes in pattern recognition – including recognizing complex stimuli like faces (the fusiform face area) and scenes. In short, the temporal lobe is a multifaceted hub for recognition and memory.

Mapping these to AI capabilities:

- **Language comprehension and production:** Human language ability relies on temporal lobe (comprehension of words, meanings) in concert with frontal lobe (speech production via Broca's area, and broader language planning). AI has seen remarkable advances here – LLMs can now parse text and generate fluent responses, indicating a high level of language competence in narrow settings. Machine translation, speech recognition, and speech synthesis are also quite advanced. Thus, for linguistic processing, AI is at L1 in many respects. An AI can "comprehend" and produce text in multiple languages with little human-like effort, though we should note it's often statistical rather than grounded understanding. Still, relative to other cognitive domains, language is a success story for AI, so the figure should mark functions like "Language Comprehension/Production" as well-developed (L1).
- **Auditory perception:** The ability to parse sound – speech, music, environmental noises – is another temporal lobe function. AI matches or exceeds humans in low-level auditory tasks like speech-to-text transcription under ideal conditions (think of virtual assistants accurately recognizing spoken commands). This is L1 for narrow cases (e.g. trained speech recognizers). However, true auditory scene analysis (understanding a cacophony of sounds, picking out one conversation in a noisy room – the "cocktail party effect" [22]) remains very hard for AI. So there are still aspects at L2/L3. But in Figure 1.1, we label "Auditory Processing (L1)" given core progress in speech recognition.
- **Episodic memory & learning:** The hippocampus enables us to form new episodic memories (remembering experiences in context) and to perform lifelong learning by integrating new memories without wiping old ones. AI's analogs here are continual learning algorithms and memory-augmented networks. This area is underdeveloped (L3) – most AI systems do not learn continuously in a stable way; they suffer catastrophic forgetting if trained on new data unless special techniques are used. They also lack the rich, context-tagged memory of experiences that humans have. We therefore label this function as "Episodic Memory & Lifelong Learning (L3)".
- **Semantic memory and understanding:** Beyond specific events, humans build up semantic memory – factual and conceptual knowledge about the world (much of this is linked to temporal lobe association areas as well). AI in some sense has simulated semantic memory: knowledge graphs, vast pretrained models that encode facts (e.g. GPT knows many facts from its training). So semantic understanding is partially there (L2). But AI's knowledge can be superficial or lacking true comprehension of context. Thus, in Figure 1.1, we include "Semantic Knowledge/Understanding (L2)" as a function.
- **Face and object recognition:** The temporal lobe's ventral stream areas identify objects and faces. AI vision is quite good at this (object and face recognition are at L1 with deep learning). This has also been captured under occipital functions already (visual perception).

Cerebellum (Coordination, Skill Learning, and Timing) The cerebellum – the "little brain" at the back – is traditionally known for motor coordination and motor learning. It fine-tunes movements, maintaining balance and posture, and ensures movements are smooth and accurate [23]. When you learn a physical skill (like riding a bicycle or playing piano), the cerebellum is heavily involved in adapting motor commands through practice – essentially performing adaptive error correction based on feedback. Notably, the cerebellum contains more neurons than the rest of the brain combined, arranged in a highly regular circuitry ideal for learning patterns. In recent decades, research has revealed the cerebellum also contributes to certain cognitive and emotional functions, acting as a predictor or timing mechanism even in non-motor tasks. It's been implicated in language (e.g. helping predict the timing of syllables) and even in aspects of attention and executive function. In essence, the cerebellum builds internal models that allow the brain to make fine-grained predictions and adjustments.

For AI, the cerebellum's roles translate to capabilities that are still not fully realized in agents:

- **Motor coordination and skill learning:** In robotics, there are control algorithms and learning methods (like reinforcement learning with feedback) that echo what the cerebellum does. For instance, adaptive controllers can learn to correct a robot arm's movements (analogous to cerebellar error correction). However, robots remain far clumsier than humans. They often lack the real-time adaptive finesse the cerebellum provides. This area is partially developed (L2) – we have examples of robotic learning for specific skills, but a general "cerebellum-like" module for adaptive motor control is not present in most AI agents. Thus, we list "Motor Skills & Coordination (L2)" by the cerebellum to highlight this gap.
- **Cognitive timing and prediction:** The cerebellum is thought to function as an internal clock and predictor for events on the order of tens to hundreds of milliseconds. This is crucial for tasks like predicting sensory outcomes of one's actions or timing when to initiate a sequence. AI systems typically do have predictive models (e.g. forward models in model-based reinforcement learning), but these are often task-specific. There isn't a general mechanism like the cerebellum's that seamlessly handles timing for perception and action across domains. Timing in AI agents (e.g. predicting when something will happen, not just what) is relatively underexplored (L3). One could imagine future AI agents with a dedicated module for temporal prediction and smooth sequencing – analogous to cerebellar function – but currently this is rudimentary.
- **Error correction:** This overlaps with coordination, but extends to cognitive domains. For example, cerebellar activity has been observed in language processing, possibly helping to predict and correct linguistic sequences [24]. AI does perform error correction in training (via backpropagation), but online error correction during tasks is limited. Real-time adaptive control (a feedback loop adjusting actions on the fly) is present in some advanced systems (e.g. adaptive cruise control in cars, or self-balancing robots), yet it's not at human proficiency. We mark "Adaptive error correction (L2)" under cerebellum to reflect that AI can do this in narrow cases but lacks a general, brain-like capability to adapt behaviors fluidly whenever mismatches occur.

The cerebellum is often left out of high-level AI comparisons, but it shouldn't be – it highlights the embodied, fine-control intelligence humans have. A particularly interesting insight from computational neuroscience is that different brain modules may correspond to different learning paradigms: cerebellum for supervised learning, basal ganglia for reinforcement learning, and cerebral cortex for unsupervised learning. This was proposed by Doya (1999) [25] and others, noting how the cerebellum takes error signals (like a supervised loss), basal ganglia use reward feedback, and cortex finds patterns in data. This perspective can inspire AI: e.g., designing an agent with separate components for these learning types, mirroring brain organization. We include the cerebellum's function (predictive control via error-driven learning) in Figure 1.1 to show that it's a major gap in current AI agents that mostly rely on reinforcement learning and (in the case of deep networks) a form of error-driven learning during training only, not continuous adaptation like the cerebellum performs in real-time.

Brainstem (Basic Autonomic and Arousal Functions) The brainstem (midbrain, pons, medulla) is the most evolutionarily ancient part of the brain, responsible for fundamental life-sustaining processes and reflexes. It acts as the main communication highway between the brain and body, and houses nuclei that control breathing, heart rate, blood pressure, swallowing, and reflexive actions like blinking [26]. In addition, the brainstem contains the reticular activating system, a network that regulates sleep-wake cycles and overall arousal level (i.e., how alert or vigilant you are). It also contributes to balance and posture (through vestibular nuclei) and coordinates head/eye movements via reflexes. Essentially, the brainstem keeps the body running and primes the brain's level of consciousness.

In terms of AI or robotics:

- **Survival autopilot:** Many of the brainstem's duties have no direct analogue in non-embodied AI (a chatbot doesn't need to regulate blood pressure!). However, in robotics, low-level control loops (for locomotion, balance, etc.) play a similar role. For instance, a bipedal robot uses feedback controllers that mimic reflexes to keep upright. These can be considered L1 (well-developed) in very narrow scopes – engineers can design reflexive responses (like a withdrawal reflex if a robot arm meets resistance). We also label "Reflexive Responses (L1)" at the brainstem, as simple reflex-like behaviors in robots (or even in software agents, e.g. an immediate reaction to an input) are straightforward and implemented.
- **Autonomic regulation:** Since AI agents don't have a body with physiology, they lack an equivalent of the autonomic nervous system. One could argue that some AI systems regulate internal variables (CPU temperature throttling, memory management) automatically, but this is a stretch as a cognitive function. Thus, we label "Autonomic Regulation (L3)" to indicate it is underexplored in AI. If we consider future embodied AI (like intelligent androids), they might need something like this to manage power, self-maintenance, etc., but it's speculative.
- **Arousal and global attention state:** The brainstem's influence on arousal has interesting parallels to AI in the sense of adaptive computation. Humans can be drowsy or hyper-alert, which affects how we process information. AI systems currently lack any explicit notion of "being alert" vs "tired" – they run in a fixed mode unless programmed otherwise. There is research into adaptive AI that could, say, slow down to save energy or limit computation when not needed, but it's not mainstream. We mark "Arousal/Attention States (L3)" as largely unaddressed. However, one might draw a parallel with how some AI models can attend more or less strongly to inputs (controlled by parameters), somewhat akin to gain control in neurons under different arousal. This is a loose analogy; overall, the global modulatory role the brainstem (with neuromodulators like norepinephrine, serotonin) plays – affecting mood and readiness – is missing in AI agents.

We should clarify that the functions we discussed here are primarily relevant for embodied AI or robotics. For instance, a self-driving car's automatic braking when a collision is imminent is reflex-like (and indeed implemented in today's tech). But an AI algorithm in isolation doesn't have "body regulation". So, brainstem functions underscore how biological intelligence is deeply tied to a body, whereas AI often abstracts that away. It's an important conceptual gap for readers to appreciate: truly human-like AI might need analogue systems for maintaining its "well-being" (self-preservation, energy management) and adjusting its alertness to situations – concepts drawn from the brainstem and related systems.

Subcortical Systems (Thalamus, Basal Ganglia, Limbic System) Finally, beyond the six major regions above, subcortical systems deserve representation, as they are crucial to cognition and differ markedly from what current AI implementations include. These structures are embedded below the cortex and often coordinate with multiple cortical regions:

- **Thalamus (Sensory Relay and Attention Filter):** The thalamus is often called the brain's "gateway" or relay station – almost all sensory signals (except smell) pass through thalamic nuclei before reaching cortex [27]. But the thalamus does more than relay: it actively modulates and integrates signals. It plays a key role in attention by amplifying relevant signals and suppressing others, under guidance from cortical feedback [28]. It's also involved in maintaining consciousness (targeted by anesthetics) and coordinating cortical rhythms. In AI, there is no single equivalent of a thalamus. However, one might liken it to routing layers or attention mechanisms that decide which data go where. The concept of a central hub that gates information flow in a network is present in some neural network architectures (e.g. transformer attention decides which inputs "attend" to which others), but the thalamus's dynamic, task-dependent control is beyond current AI. We mark functions like "Sensory Integration & Routing" as L2 (partially present conceptually in AI via attention layers), and "Global Workspace" as L3 (largely absent – AI doesn't have a unified workspace model equivalent to what some cognitive theories assign to thalamocortical circuits).

- **Basal Ganglia (Action Selection and Reinforcement Learning):** The basal ganglia are a group of nuclei (caudate, putamen, globus pallidus, substantia nigra, etc.) that are central to selecting and initiating actions, and they implement a biological form of reinforcement learning. They take inputs from the cortex (especially frontal and parietal areas), and through complex loops, they determine which actions are facilitated or inhibited, often by evaluating expected rewards or outcomes [16]. Dopamine signals from the midbrain (e.g. from the substantia nigra pars compacta) encode reward prediction errors, a concept very much like the reward signals in AI RL algorithms. In fact, neuroscientific evidence suggests the basal ganglia "learn" which actions lead to reward via dopamine-mediated plasticity – a direct parallel to how AI agents update policies from reward feedback. We can confidently say "Reward-Based Learning and Habit Formation" are primary functions of the basal ganglia. AI has a whole subfield of reinforcement learning, which has seen successes (games, some robotic tasks), so this is moderately developed (L2) in AI. However, current AI RL is narrow and data-hungry compared to human habit learning. Basal ganglia also contribute to procedural memory (learning habits or skills that become automatic) – something AI doesn't explicitly differentiate (it learns policies, but the idea of habits vs. goal-directed actions is an emerging concept in AI research).

- **Limbic System – Amygdala and Hippocampus (Emotion and Memory):** We touched on these in the temporal lobe section, but to reiterate: the amygdala is crucial for processing emotional significance of stimuli and fear conditioning (learning to avoid harmful situations) [29]. It assigns value (good or bad) to experiences, which then influences decision-making and memory (through its connections to hippocampus and frontal cortex). The hippocampus, as mentioned, enables forming new declarative memories and mapping environments (it's often likened to the brain's GPS for spatial memory). In AI, there are nascent attempts to model hippocampal function – e.g. neural network "memory" modules for episodic recall, or models of spatial navigation that emulate place cells and grid cells found in the hippocampal formation. These are still L3 overall (exploratory). As for the amygdala's role, AI currently lacks genuine emotion; at most, we simulate "emotion" as reward functions or use sentiment analysis to detect emotions in text, which is not an internal drive. We label "Emotion Processiong & Learning (L3)" and "Episodic Memory (L3)" to highlight capabilities largely missing in AI agents: affective computing (AI that can experience or at least robustly respond to emotions) is very limited, and one-shot contextual learning (storing a new event and generalizing from it) is also an open problem.

- **Hypothalamus (Drives and Homeostasis):** The hypothalamus orchestrates the endocrine system and autonomic nervous system to maintain internal balance – it controls things like hunger, thirst, temperature, and release of hormones [30]. It also generates primitive drives (e.g. hunger drives you to seek food, which the cortex then plans for). AI agents do not have intrinsic survival needs, so they lack any true equivalent of homeostatic drives. We sometimes give AI an objective function (e.g. maximize score), but these are externally defined and do not fluctuate like biological needs. To the extent researchers are exploring intrinsic motivation for AI (like curiosity-based rewards), it is still rudimentary. In the figure, we add "Motivation & Drives (L3)" to acknowledge this gap. It reminds us that a human-inspired AI agent might require internally generated goals (not just tasks imposed by users) to be truly autonomous and robust in varied environments. It should be noted that giving AI intrinsic motivation is also seen as a potentially dangerous direction and should be treated with great caution [31].

Bringing these subcortical pieces together, we see that many are poorly represented in today's AI. Current intelligent systems are heavily cortex-like (perception modules, decision logic, etc.) but lack the rich support system of the subcortical brain: no analog of a thalamus to smartly route information, no hypothalamus to create self-preserving goals, a primitive version of basal ganglia for RL at best, and minimal emotional or episodic memory faculties. The figure's accompanying explanation should drive home that cognition arises from cortical–subcortical interactions. For example, decision-making is not just frontal (cortical) deliberation, but also involves basal ganglia (habits and dopamine rewards) and amygdala (emotional bias) and hypothalamus (drive states). Emphasizing these interactions will lend a more nuanced and truthful picture than a simplistic lobe-by-lobe map. It also inspires AI researchers to think about architectures that incorporate these principles – such as an agent that, say, has a core RL module (analogous to basal ganglia) for learning from rewards, a memory module (analogous to hippocampus) for episodic recall, and perhaps a "global workspace" (inspired by thalamocortical loops) for attention and context integration.

Bridging Brain-Like Functions and Building Beneficial AI Until now, we have witnessed the gap between human brain and machine intelligence. Nevertheless, the objective is not necessarily to replicate every facet of human cognition within artificial intelligence systems. Rather, our overarching aim should be to develop intelligent agents that are useful, ethical, safe, and beneficial to society. By critically comparing human and artificial intelligence, we highlight the existing gaps and illuminate promising directions for innovation. This comparative perspective allows us to selectively integrate beneficial aspects of human cognition, such as energy-efficient processing, lifelong adaptive learning, emotional grounding, and rich creativity, while simultaneously innovating beyond human limitations. Ultimately, this approach aims to foster the creation of more capable, resilient, and responsible AI systems.

Furthermore, it is vital to consider the evolving role of humans within a hybrid Human-AI society. The goal of AI should not be to replace human roles entirely, but rather to augment and empower human abilities, complementing human skills and judgment in areas where AI excels, such as handling vast datasets, performing rapid calculations, and automating repetitive tasks. Human oversight and interpretability are essential to ensure that powerful AI systems remain controllable and aligned with human values and ethical standards. Thus, the core objective must be the development of AI technologies that are transparent, interpretable, and responsive to human guidance.

Human-centered AI design emphasizes collaboration, safety, and social responsibility, ensuring technological advancement proceeds in a controlled, reliable manner. By placing humans at the center of the AI ecosystem, we can harness AI's potential to enhance human productivity, creativity, and decision-making, facilitating technical and societal progress without compromising human autonomy or dignity. Ultimately, a thoughtful integration of human intelligence and AI capabilities can pave the way for a sustainable, equitable, and prosperous future.

PART II

中文译文

Chinese Translation

对应第 1.2 节与 1.2.1 节全文

1.2 人脑与 AI 智能体的平行比较

大语言模型（LLM）迅速融入智能体架构，不仅推动了人工智能的发展，也凸显出 AI 系统与人类认知之间的根本差异。如表 1.1 所作的简要展示，LLM 驱动的智能体在人机底层“硬件”、意识、学习机制、创造力和能源效率等维度上，都与人类认知存在显著不同。不过需要强调的是，这一比较只是高层次概览，并非穷尽式描述。人类智能包含许多此处未能涵盖的细微特征，AI 智能体也拥有超出这份简要比较的独特属性。

人类智能运行在生物硬件 - 大脑 - 之上。大脑具有非凡的能源效率，能以很低的代谢成本完成终身学习、推断和适应性决策。相比之下，当前 AI 系统需要大量算力，在完成相当的认知任务时耗能显著更高。认识到这一性能差距，意味着能源效率应成为未来 AI 研究的重要前沿。

人类学习是持续的、互动的、对情境敏感的，并深受社会、文化和经验因素塑造。相反，LLM 智能体主要接受静态的离线批量训练，持续适应能力有限。尽管已有指令微调和基于人类反馈的强化学习（RLHF）研究 [12]，LLM 智能体仍未达到人类式的灵活性。通过终身学习、个性化适应和交互式微调等方法弥合这一差距，是很有前景的研究方向，可使 AI 更接近人类的适应性与响应能力。

人类创造力源自个人经历、情感洞见和自发跨领域联想之间的丰富互动。情绪状态不仅会推动创造性表达，还会影响成果的原创性、深度与感染力，使其具有个人意义和情感分量。相比之下，大语言模型的创造力主要来自对训练数据的统计重组，可称为“统计式创造力”。这种创造力往往流畅，有时也令人意外，但缺少情感根基、亲身经历与有意图的原创性。这种差异提示了一些推进 AI 智能体的机会：纳入更深入的情境理解、模拟情绪状态和经验性记忆，从而形成更真实、对情感更敏锐的创造过程。

在意识与情感体验方面，LLM 智能体缺乏人类认知所具有的真实主观状态和自我意识。让 AI 完整复制人类式意识或许既无必要，也未必可取；但理解情感和主观体验在人类推理、动机、伦理判断与社会交往中的深刻作用，可以引导研究者构建更对齐、更可信、对社会更有益的 AI。

从时间尺度看，人脑经过数百万年演化，在自然选择和环境互动中获得了卓越的效率、适应性与创造力。与之形成鲜明对照的是，自早期计算机器出现以来，AI 智能体只经历了大约 80 年的快速而短暂的发展。因此，将人类认知与 AI 系统作平行比较很有价值：它能揭示根本差异，并为 AI 智能体技术进步提供有意义的启示。最终，从人类智能汲取灵感可以增强 AI 能力，并在医疗、教育、可持续发展等广泛领域造福人类。

表 1.1 人脑与 LLM 智能体的简明高层比较（中文重排）

硬件与维护	人脑 / 认知	LLM 智能体
说明	• 生物神经元、神经递质与神经可塑性。 • 需要睡眠、营养和休息。 • 难以复制，知识通过学习传递。 • 能源效率极高（约 20W）。	• 深度神经网络与基于梯度的优化。 • 需要硬件、稳定供电与冷却。 • 可在全球服务器间轻易复制。 • 高能耗（每台 GPU 服务器数千瓦）。
比较结论	人脑由生物机制维持，能源效率高且不易复制；LLM 智能体依赖硬件维护，复制性很强，但能源效率显著更低。	

学习方式	人脑 / 认知	LLM 智能体
说明	• 终身、持续、在线学习。 • 少样本、快速知识迁移。 • 受环境、文化和情感影响。	• 主要是离线、批量训练。 • 在线学习与适应能力有限。 • 学得的知识中性且非人格化。
比较结论	尽管指令微调带来了改进，人类学习仍比 LLM 学习更动态、更具适应性，也与文化和情感结合得更深。	

创造力与发散	人脑 / 认知	LLM 智能体
说明	• 根植于个人经历、情感和潜意识洞见。 • 具有丰富的跨领域联想和隐喻思维。 • 情感深度影响创造力。	• 从海量数据中进行统计重组。 • 通过概率优化产生新颖性。 • 情感和经验根基有限。
比较结论	LLM 的创造力是统计式、数据驱动的；人类创造力则融合了情感、经验和潜意识过程。	

意识与发展	人脑 / 认知	LLM 智能体
说明	• 真实的主观体验、情感与自我意识。 • 从童年开始经历渐进的发展阶段。 • 情感认知驱动决策。	• 没有真实主观体验或自我意识。 • “情感”只是表层语言模仿。 • 训练后基本静态，动态成长有限。
比较结论	人类意识从情感、社会和生物发展中涌现；LLM 仍然静态，缺少真正的自省和情感深度。	

1.2.1 脑功能与 AI 对照

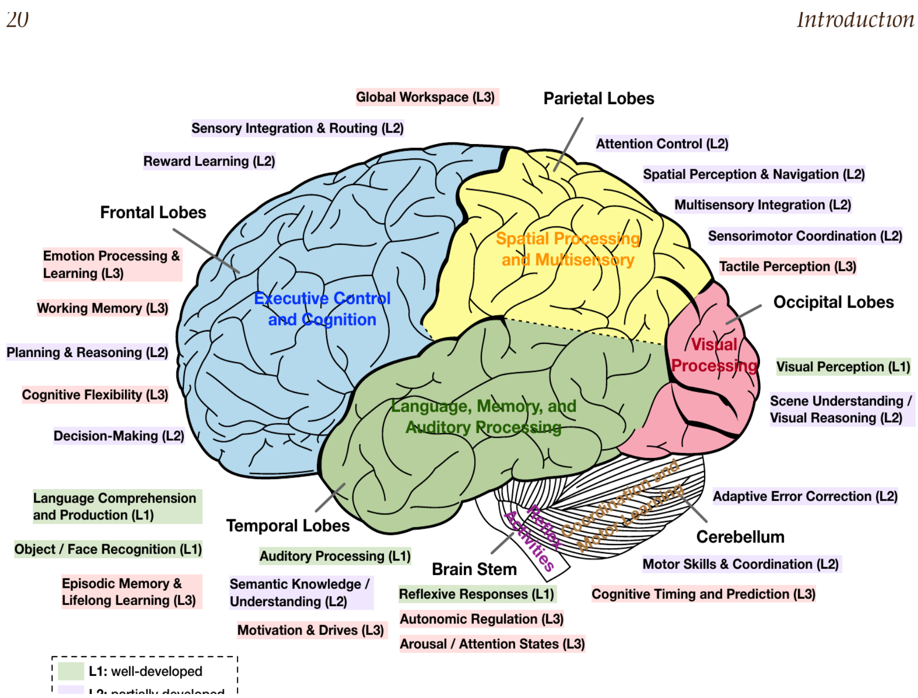
设计智能体需要从人脑的功能架构中汲取灵感。把额叶、顶叶、枕叶、颞叶、小脑、脑干及关键皮层下结构与认知功能相联系的高层地图，可以揭示人类能力与当前 AI 系统之间的差距，如图 1.1 所示。不过，脑功能并非被封闭在刚性的解剖区域内；多数能力来自跨越多个脑区的网络。例如，记忆涉及海马体（颞叶）与额叶皮层及其他区域的互动，而“自我意识”或意识也无法被定位到某一个点。因此必须牢记：认知是分布式的，并非严格局部化。

基于这一认识，作者回顾各主要脑区的核心功能（参考 Kandel 等人的《神经科学原理》[13]、Purves 等人的《神经科学》[14] 及其他资料），并将其映射为与 AI 相关的认知能力。图中选取了与 AI 智能体最相关的功能，包括推理、记忆、规划、感知、决策、动机、情感和运动技能，并评估这些能力在 AI 中的发展程度。为形成总体视角，AI 研究现状被划分为三个层级：

- 第 1 级（L1）：发展成熟。
- 第 2 级（L2）：部分发展。
- 第 3 级（L3）：探索不足。

目标是形成一幅清晰、具有生物学依据的图示与讨论：指出哪些人类认知功能已在机器中得到复制，哪些仍是前沿挑战，从而吸引并启发 AI 研究者。

图 1.1 不同脑功能及其在 AI 中的研究状态（2025）



图示按主要脑区归类人脑的关键功能，并标注其在当前 AI 研究中的探索层级，突出既有成果、能力缺口以及推动更全面的脑启发式人工智能的潜在机会。标签所在位置不意味着该功能只与邻近脑区相关。为了聚焦 AI 能力现状而非脑解剖结构，原图的侧面脑视图未标出边缘系统、皮层下系统、下丘脑、杏仁核等结构，尽管它们对脑功能十分重要。

脑区 / 系统	功能标签（中文）
额叶	执行控制与认知；规划与推理（L2）；决策（L2）；工作记忆（L3）；认知灵活性（L3）；情感处理与学习（L3）
顶叶	空间处理与多感觉整合；注意控制（L2）；空间感知与导航（L2）；多感觉整合（L2）；感觉运动协调（L2）；触觉感知（L3）
枕叶	视觉处理；视觉感知（L1）；场景理解 / 视觉推理（L2）
颞叶	语言、记忆与听觉处理；语言理解与生成（L1）；听觉处理（L1）；物体 / 人脸识别（L1）；语义知识 / 理解（L2）；情景记忆与终身学习（L3）
小脑	运动协调与运动学习；运动技能与协调（L2）；自适应误差校正（L2）；认知计时与预测（L3）
脑干 / 相关系统	反射活动；反射式响应（L1）；自主调节（L3）；唤醒 / 注意状态（L3）；感觉整合与路由（L2）；奖励学习（L2）；全局工作空间（L3）；动机与驱力（L3）

额叶（执行功能与决策）

额叶 - 尤其是前额叶皮层 - 是大脑最高阶认知功能的核心，这些功能统称为执行功能 [13]，包括规划、决策、问题求解、工作记忆和抑制控制（自我控制）。额叶还参与随意运动控制（后部含初级运动皮层）和语言功能（左额叶的布罗卡区负责言语生成）。从神经科学看，前额叶受损会显著损害判断、规划和社会行为，经典的菲尼亚斯·盖奇病例就是例证 [15]。在 AI 智能体中，额叶功能对应智能系统的核心“思考”和控制部件：

- 规划与推理：AI 已在自动规划器、逻辑推理器以及能够在一定程度上执行多步推理的 LLM 方面取得进展。当前属于部分发展（L2），但人类水平的灵活规划仍只得到部分解决。

- 决策：人类决策会权衡结果、奖励和风险，额叶皮层常与基底神经节协作 [16]。AI 智能体拥有强化学习策略、决策树或 LLM 等决策模块，但面对开放式、目标冲突的决策时，其适应性充其量仍为 L2。AI 已掌握奖励定义明确的简单决策（如游戏），而现实世界中的广泛自主决策仍很困难。

- 工作记忆：额叶网络，特别是背外侧前额叶皮层，可在头脑中保持并操纵信息，例如记住电话号码或中间结果。AI 中的对应物包括神经网络上下文窗口和显式记忆缓冲区。当前模型的记忆有限（如 Transformer 的上下文长度，或某些架构中的外部记忆），这是一个活跃研究领域，也已经具备部分功能；但人类那种稳健、通用、可灵活更新并聚焦相关信息的工作记忆，尚未在 AI 中充分实现，其中一些方面接近探索不足的 L3。

- 认知灵活性与抑制控制：额叶使人能够转换策略或视角，并抑制不恰当的冲动。AI 在这方面通常很脆弱：它会僵化地遵循程序或已学策略，除非专门训练，否则难以临场改变策略或抑制优势反应。这仍属 L3。例如，如果设计者没有预见并约束，AI 可能钻奖励漏洞，表现出缺少"抑制性"自我调节。未来智能体架构或许需要类似额叶抑制控制的机制来调节行为。

还应注意，涉及额叶的社会情感功能，如同理心、心智理论和自我反思，在 AI 中仍很初步。人类依靠额叶皮层，尤其是内侧和眶额区域，与边缘结构互动，来处理社会情境和情绪。这是网络层面的现象，不只局限于额叶。AI 智能体尚不具备真实同理心或自我意识，这些功能仍为 L3（基本缺失）。总之，额叶提供了通常被称为"思考"的监督性、目标导向智能，而其中许多能力迄今只在 AI 中得到部分实现。

顶叶（感知整合与注意）

顶叶对于整合不同感觉模态的信息并构建对世界的空间理解至关重要 [14]。顶叶前部含躯体感觉皮层，负责处理触觉、本体感觉（身体位置）和其他躯体感觉。顶叶后部对空间意识、视觉空间处理和注意极为关键，也就是理解事物"在哪里"以及如何与之互动。例如，顶叶帮助我们在空间中定位物体、理解几何关系，并把视觉与运动计划联系起来以协调眼手动作。它还在注意控制中起核心作用，尤其是背侧注意网络会把注意导向感兴趣的位置或感觉特征。换成 AI 术语，顶叶功能对应智能体感知和导航环境的能力：

- 多感觉整合：使用视觉、触觉等多种传感器的机器人和 AI 系统，会尝试把输入合成为连贯的环境模型。这仍是 L2。AI 可以对齐视觉、深度传感器或触觉，但距离人类水平的整合还很远；人可以把手臂上的轻触、声音与周边视野中的线索统一为一个事件。

- 空间表征与建图：顶叶回路会建立内部地图，例如周围环境或身体在空间中的地图。AI 在空间建图和导航上已有进展：机器人用同步定位与建图（SLAM）算法构建三维地图，深度强化学习智能体也能在虚拟迷宫中导航。自动驾驶或无人机飞行说明机器可在受限场景中进行空间推理，因此为 L2。但它们缺少人类通用且灵活的空间理解，例如一眼把握杂乱房间的布局或在头脑中旋转物体。

- 注意机制：大脑顶叶-额叶回路突出任务相关信息的方式，与现代 Transformer 的注意头 [17] 和各种注意机制 [18] 存在宽泛对应。但人类把注意当作主动镜头：听讲时可同时跟随讲者声音、浏览投影并用周边视野看时钟；随后又能"放大"一行文字来解析公式，或"缩小"以把握演讲整体结构。神经生理研究表明，这种快速切换由前额叶皮层和丘脑中继发出的自上而下信号驱动，能即时调节感觉增益 [19]。相比之下，Transformer 的注意在权重学习完成后是固定的，不能获得有关目标或情境的实时执行反馈。因此当前 AI 被标为 L2：计算注意很强，但仍缺少生物注意那种适应性、目标导向的控制。

- 感觉运动协调：顶叶帮助在感觉坐标与运动坐标之间转换，例如把视觉位置与手臂位置结合，计算如何伸手拿到看见的物体。配备视觉的机械臂可利用标定和学习得到的坐标变换近似实现这一点；但人类顶叶擅长在线调整和利用情境，例如目标移动时修正伸手动作。AI 在机械臂操控等领域正迎头赶上，但通用感觉运动整合仍为 L2，只在特定装置中得到证明，稳健性不如人类。

尤其是，AI 的触觉感知十分有限，因此属于 L3；而顶叶躯体感觉皮层能精细区分质地、压力等属性。总体而言，顶叶功能对于智能体感知世界并在其中确定方位至关重要。AI 在这些领域已有一定成功，尤其是视觉，但仍缺少人类感知的通用性与流畅性。

枕叶（视觉处理）

枕叶是大脑的视觉处理中心 [13]。枕叶的初级视觉皮层（V1）经丘脑接收眼睛输入，并提取边缘等低层特征。随后，枕叶区域和邻近视觉区域沿腹侧通路延伸至枕颞交界、沿背侧通路延伸至枕顶区域，分层构建视觉知觉：先检测形状、颜色和运动，再识别复杂模式和物体。简言之，枕叶主要负责处理视觉信息。在 AI 中，得益于深度学习，视觉是最成功的领域之一：

- 视觉感知（识别）：机器视觉系统（卷积神经网络及其后继模型）在物体识别、人脸检测和图像分类等任务上已经达到或超过人类表现 [20]，对应发展成熟的 L1。例如，AI 视觉模型可以立刻识别图像中的数千种物体类别，这一能力曾被认为需要人类式视觉。它对应枕叶及腹侧视觉皮层识别视野中“是什么”的功能。
- 场景理解与视觉推理：除了原始感知，人类能轻松理解场景中的空间关系和情境线索，并对视觉输入推理，例如预测场景接下来会发生什么或解决视觉谜题。AI 在这里仅达到 L2。一些视觉-语言模型能为图像生成说明或回答场景问题，显示出一定语义理解；但它们往往缺少真正扎根现实的理解，可能会标注物体，却无法理解图像中的意图或因果。回答图片抽象问题、复杂视频分析等视觉推理任务仍然困难。
- 视觉注意与眼动：人类不断移动眼睛并聚焦视野中的重要部分，这涉及枕叶和顶叶回路。AI 视觉模型不会真的移动眼睛，但有些模型包含模拟聚焦图像区域的注意机制。这与前述注意功能相关，属于顶叶和额叶共享的协调功能。AI 在该方面约为 L2；原图未把它单列在枕叶下，而将其视为一般注意功能的一部分。

AI 已高度发展视觉识别能力，但实时三维视觉引导仍不够成熟，尽管机器人和自动驾驶汽车中已经有所应用。这一区别体现了 AI 与人类视觉系统高度整合能力之间的又一差距。

颞叶（记忆、语言与听觉）

颞叶在认知中承担多样而关键的作用，涵盖听觉处理、语言、记忆和高层视觉识别 [14]。颞叶上部的颞上回包含初级听觉皮层，处理频率、节奏等声音输入。邻近区域，如左颞叶的韦尼克区 [21]，对语言理解至关重要，负责把声音与意义相联系。内侧颞叶包含海马体及相关结构，是大脑情景记忆系统的核心，负责形成和提取自传体记忆，也支持空间导航。颞叶腹侧视觉通路（下颞叶皮层）专长于模式识别，包括人脸（梭状回面孔区）、场景等复杂刺激。简言之，颞叶是识别和记忆的多功能枢纽。映射到 AI 能力如下：

- 语言理解与生成：人类语言能力依赖颞叶理解词语和意义，并与额叶协同：布罗卡区负责言语生成，额叶还参与更广泛的语言规划。AI 在此取得了显著进展，LLM 可解析文本并生成流畅回答，在有限场景中表现出较高语言能力；机器翻译、语音识别和语音合成也相当先进。因此 AI 的语言处理在许多方面达到 L1。AI 能以很少的人类式努力“理解”和生成多种语言的文本，不过这种理解通常是统计性的，而非扎根现实。相较其他认知领域，语言仍是 AI 的成功案例，图中将“语言理解与生成”标为 L1。
- 听觉感知：解析语言、音乐和环境噪声等声音也是颞叶功能。在理想条件下，AI 在语音转文字等低层听觉任务上可达到或超过人类，例如虚拟助手能准确识别口头命令；在训练良好的窄域语音识别中属于 L1。但真正的听觉场景分析 - 理解嘈杂声场，或在吵闹房间中分离一段对话，即“鸡尾酒会效应” [22] - 对 AI 仍然很难，部分方面属于 L2/L3。鉴于语音识别的核心进展，图 1.1 仍把“听觉处理”标为 L1。
- 情景记忆与学习：海马体使人形成新的情景记忆，在情境中记住经历，并通过整合新记忆而不抹去旧记忆来实现终身学习。AI 的对应方法包括持续学习算法和记忆增强网络。该领域仍不成熟（L3）：多数 AI 系统无法稳定地持续学习，除非使用特殊技术，否则在新数据上训练会发生灾难性遗忘；它们也缺少人类那种带有丰富情境标签的经验记忆。因此图中标为“情景记忆与终身学习（L3）”。

- 语义记忆与理解：除了具体事件，人类还会积累关于世界的事实性、概念性知识，即语义记忆，其中许多与颞叶联络区有关。从某种意义上说，AI 已模拟出语义记忆：知识图谱和海量预训练模型都能编码事实，例如 GPT 从训练中获知许多事实。因此语义理解部分存在（L2），但 AI 的知识可能很表层，或缺少对情境的真正理解。图中据此列出“语义知识 / 理解（L2）”。
- 人脸与物体识别：颞叶腹侧通路负责识别人脸和物体。AI 视觉在此表现很好，深度学习的人脸与物体识别已达到 L1；这一能力也已归入枕叶功能中的视觉感知。

小脑（协调、技能学习与计时）

位于脑后部的小脑传统上以运动协调和运动学习著称。它会微调运动、维持平衡与姿势，使动作平稳准确 [23]。学习骑车或弹琴等身体技能时，小脑通过练习不断调整运动指令，本质上是基于反馈执行自适应误差校正。值得注意的是，小脑所含神经元数量超过大脑其余部分的总和，并以高度规则、适合学习模式的回路排列。近几十年的研究表明，小脑也参与某些认知和情绪功能，即便在非运动任务中也可充当预测器或计时机制；它与语言（如预测音节时序）、注意和执行功能都有关系。小脑会构建内部模型，使大脑能够做出细粒度预测和调整。其功能在 AI 智能体中尚未得到充分实现：

- 运动协调与技能学习：机器人控制算法和带反馈的强化学习等方法与小脑功能相呼应。例如，自适应控制器可学习修正机械臂运动，类似小脑误差校正。但机器人仍比人类笨拙，通常缺少小脑提供的实时适应性精细控制。该领域为 L2：已有针对特定技能的机器人学习，但多数 AI 智能体没有通用的“小脑式”自适应运动控制模块，因此图中列为“运动技能与协调（L2）”。
- 认知计时与预测：小脑被认为是内部时钟和预测器，处理几十到几百毫秒尺度的事件；这对于预测自身动作的感觉结果，或决定何时启动一个序列至关重要。AI 系统通常也有预测模型，如基于模型的强化学习中的前向模型，但往往只针对特定任务。AI 缺少像小脑那样跨领域无缝处理感知与动作计时的通用机制。智能体对“何时发生”而不只是“发生什么”的预测仍属 L3。未来可以设想专门负责时间预测和平滑排序、类似小脑的模块，但目前还很初步。
- 误差校正：该功能与协调重叠，也延伸至认知领域。语言处理中可以观察到小脑活动，它可能帮助预测并校正语言序列 [24]。AI 训练会通过反向传播校正误差，但任务执行时的在线误差校正有限。一些先进系统具有实时自适应控制，如汽车自适应巡航或自平衡机器人，但仍未达到人类熟练度。因此“小脑自适应误差校正”标为 L2：AI 可在狭窄场景中做到，却缺少在出现不匹配时流畅调整行为的通用脑式能力。

高层 AI 比较经常忽略小脑，但不应如此，因为它体现了人类具身、精细控制的智能。计算神经科学有一项特别有趣的洞见：不同脑模块或许对应不同学习范式 - 小脑对应监督学习，基底神经节对应强化学习，大脑皮层对应无监督学习。Doya（1999）[25] 等人提出，小脑接收类似监督损失的误差信号，基底神经节使用奖励反馈，皮层则从数据中发现模式。这可以启发 AI，例如设计分别负责这些学习类型的组件，模拟脑组织。图 1.1 纳入小脑通过误差驱动学习进行预测控制的功能，是为了显示当前 AI 智能体的一项主要缺口：它们主要依靠强化学习，深度网络则只在训练阶段执行某种误差驱动学习，而没有像小脑那样实时持续适应。

脑干（基本自主功能与唤醒）

脑干由中脑、脑桥和延髓构成，是大脑在演化上最古老的部分，负责维持生命的基本过程与反射。它是脑与身体之间的主要通信通道，包含控制呼吸、心率、血压、吞咽和眨眼等反射动作的神经核 [26]。脑干还包含网状激活系统，负责调节睡眠-觉醒周期与整体唤醒水平，也就是人的清醒或警觉程度。它还通过前庭核参与平衡与姿势，并通过反射协调头眼运动。实质上，脑干让身体持续运转，并调节大脑的意识水平。对应到 AI 或机器人：

- 生存自动驾驶：许多脑干职责在非具身 AI 中没有直接对应物，聊天机器人不需要调节血压。但在机器人中，行走、平衡等低层控制回路扮演类似角色。例如，双足机器人用模拟反射的反馈控制器保持站立。在很窄的范围内，这些能力可视为 L1：工程师能设计反射式响应，如机械臂遇阻时立即回撤。图中也把“反射式响应”标为 L1，因为机器人乃至软件智能体的简单即时反应都容易实现。

- 自主调节：AI 智能体没有具有生理机能的身体，因此缺少自主神经系统的对应物。有人可能认为 CPU 温度节流、内存管理等自动调节内部变量的机制与之类似，但若把它们视为认知功能就相当牵强。因此“自主调节”被标为 L3。未来的具身 AI，如智能仿生人，可能需要类似机制来管理供能和自我维护，但这仍属推测。
- 唤醒与全局注意状态：从适应性计算角度看，脑干对唤醒的影响与 AI 具有有趣的类比。人会困倦或高度警觉，这会改变信息处理方式；AI 则没有明确的“清醒”或“疲劳”概念，除非被编程，否则总以固定模式运行。已有自适应 AI 研究尝试在不需要时减速节能或限制计算，但尚非主流。因此“唤醒 / 注意状态”标为 L3。某些 AI 模型可通过参数对输入给予不同强度的注意，这与不同唤醒状态下神经元的增益控制有松散类比；但总体而言，脑干借助去甲肾上腺素、血清素等神经调质对情绪和准备状态实施的全局调节，在 AI 智能体中仍然缺失。

这里讨论的功能主要与具身 AI 或机器人有关。例如，自动驾驶汽车在即将碰撞时自动刹车，就是一种已在现有技术中实现的反射式行为；但孤立的 AI 算法并不具有“身体调节”。因此，脑干功能凸显出生物智能与身体的深度联系，而 AI 往往把身体抽象掉。这个概念差距值得重视：真正类人的 AI 可能需要维持自身“福祉”（自我保存、能源管理）并根据情境调整警觉性的对应系统，这些概念来自脑干及相关系统。

皮层下系统（丘脑、基底神经节与边缘系统）

除上述六个主要区域外，皮层下系统也应得到呈现，因为它们对认知至关重要，并与当前 AI 实现显著不同。这些结构位于皮层下方，常与多个皮层区域协同：

- 丘脑（感觉中继与注意过滤器）：丘脑常被称为大脑的“门户”或中继站，除嗅觉外，几乎所有感觉信号在到达皮层前都要经过丘脑核 [27]。但丘脑不只负责中继，它还主动调制和整合信号，在皮层反馈指导下放大相关信号、抑制无关信号，因此在注意中发挥关键作用 [28]；它也参与维持意识（麻醉药会作用于此）和协调皮层节律。AI 中没有单一的丘脑对应物，不过可把决定数据去向的路由层或注意机制与之类比。有些神经网络架构已有控制信息流的中心枢纽概念，如 Transformer 注意决定哪些输入彼此“关注”，但丘脑动态、依任务而变的控制超出了当前 AI。图中把“感觉整合与路由”标为 L2，把“全局工作空间”标为 L3，因为 AI 基本没有认知理论赋予丘脑-皮层回路的统一工作空间模型。
- 基底神经节（动作选择与强化学习）：基底神经节由尾状核、壳核、苍白球、黑质等神经核组成，是动作选择和启动的核心，并实现一种生物学形式的强化学习。它接收皮层输入，尤其来自额叶与顶叶的输入，经复杂回路判断应促进或抑制哪些动作，常以预期奖励或结果为依据 [16]。中脑多巴胺信号，例如来自黑质致密部的信号，会编码奖励预测误差，这与 AI 强化学习算法中的奖励信号非常相似。神经科学证据表明，基底神经节通过多巴胺介导的可塑性“学习”哪些动作能带来奖励，直接对应 AI 智能体如何根据奖励反馈更新策略。因此，“基于奖励的学习与习惯形成”可视为基底神经节的主要功能。AI 的强化学习子领域已在游戏和部分机器人任务中取得成功，因而达到 L2；但与人类习惯学习相比，当前 AI 强化学习范围狭窄且耗费数据。基底神经节还参与程序性记忆，即学习逐渐自动化的习惯或技能；AI 并未明确区分这一点，虽然它会学习策略，但习惯与目标导向动作之分仍是新兴研究概念。
- 边缘系统 - 杏仁核与海马体（情感与记忆）：这些结构在颞叶部分已经提到。杏仁核对处理刺激的情感意义和恐惧条件化至关重要，可学习规避有害情境 [29]。它给经历赋予好坏价值，再通过与海马体和额叶皮层的连接影响决策和记忆。海马体负责形成新的陈述性记忆和绘制环境地图，常被比作负责空间记忆的“大脑 GPS”。AI 已有模拟海马体功能的初步尝试，例如用于情景回忆的神经网络“记忆”模块，或模拟海马结构中位置细胞与网格细胞的空间导航模型；整体仍属探索性的 L3。至于杏仁核，AI 没有真实情感，最多把“情感”模拟为奖励函数，或用情感分析检测文本中的情绪，但这不是内在驱力。因此图中把“情感处理与学习”和“情景记忆”都标为 L3，以突出 AI 智能体中基本缺失的能力：能体验情感或至少稳健响应情感的情感计算非常有限，一次性情境学习 - 保存一个新事件并据此泛化 - 也仍是开放问题。
- 下丘脑（驱力与稳态）：下丘脑协调内分泌系统和自主神经系统以维持内部平衡，控制饥饿、口渴、体温和激素释放等 [30]；它也会产生原始驱力，例如饥饿驱使人寻找食物，再由皮层制定计划。AI 智能体没有内在生存需求，因此没有真正的稳态驱力对应物。人们有时给 AI 设定目标函数，如最大化分数，但这些目标由外部定义，并不会像生物

需要那样波动。即使研究者开始探索好奇心奖励等 AI 内在动机，也仍很初步。图中加入“动机与驱力（L3）”来体现这一缺口：若要在多样环境中真正自主而稳健，脑启发式 AI 智能体或许需要内部生成目标，而不只是执行用户施加的任务。同时必须指出，赋予 AI 内在动机也被认为可能带来危险，应极其谨慎 [31]。

综合这些皮层下组成可以看到，它们在今天的 AI 中大多没有得到良好呈现。当前智能系统高度“类皮层”，拥有感知模块、决策逻辑等，却缺少皮层下大脑丰富的支持系统：没有丘脑对应物来智能路由信息，没有下丘脑产生自我保存目标，最多只有用于强化学习的原始“基底神经节”，情感能力和情景记忆也极少。图示的说明应强调，认知来自皮层与皮层下结构的互动。例如，决策不只是额叶皮层的审议，还涉及基底神经节中的习惯和多巴胺奖励、杏仁核的情绪偏向及下丘脑的驱力状态。强调这些互动，比逐脑叶的简单地图更细致、更真实；也能启发 AI 研究者设计体现这些原则的架构，例如用类似基底神经节的核心强化学习模块从奖励中学习，用类似海马体的记忆模块进行情景回忆，再用受丘脑-皮层回路启发的“全局工作空间”整合注意与情境。

衔接类脑功能，构建有益 AI

到目前为止，我们已经看到人脑与机器智能之间的差距。不过，目标未必是在人工智能系统中复制人类认知的每一个方面；更高层的目标应当是开发有用、合乎伦理、安全并有益于社会的智能体。通过批判性比较人类智能与人工智能，我们可以揭示现有缺口，照亮有前景的创新方向。这种比较视角使我们能够选择性吸收人类认知的有益方面，如高能效处理、终身适应性学习、情感根基和丰富创造力，同时又超越人类局限进行创新。最终，这一路径旨在促进更有能力、更具韧性、更负责任的 AI 系统。

此外，必须考虑人在混合人机社会中不断演变的角色。AI 的目标不应是彻底取代人类角色，而应增强并赋能人的能力：在 AI 擅长的领域，例如处理海量数据、快速计算和自动化重复任务，补充人类技能与判断。人类监督和可解释性不可或缺，它们能确保强大的 AI 系统仍然可控，并与人类价值和伦理标准对齐。因此，核心目标必须是发展透明、可解释、能响应人类指导的 AI 技术。

以人为中心的 AI 设计强调协作、安全和社会责任，确保技术进步以受控、可靠的方式进行。把人置于 AI 生态系统中心，可以利用 AI 提升人的生产力、创造力和决策能力，在不损害人类自主与尊严的前提下推动技术和社会进步。最终，审慎融合人类智能与 AI 能力，将为可持续、公平而繁荣的未来铺平道路。

引用提示

本译文保留原文引用编号 [12]-[31]。完整参考文献及上下文请参阅原论文 Advances and Challenges in Foundation Agents: From Brain-Inspired Intelligence to Evolutionary, Collaborative, and Safe Systems（arXiv:2504.01990v2）。